

JK BMS Monitor

User Manual

Black Sphere Industries

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Table of Contents

1	Introduction	3
2	Getting Started	3
2.1	System Requirements	3
2.2	Accessing the Application	3
2.3	Connecting a BMS	3
3	Features	4
3.1	Multi-Device Monitoring	4
3.2	Battery Overview	4
3.3	Cell Voltage Monitoring	4
3.4	Temperature Monitoring	4
3.5	MOSFET and Protection Status	4
3.6	Error Detection	5
3.7	Device Information	5
3.8	Debug Tools	5
3.8.1	Hex Dump	5
3.8.2	Activity Log	5
3.9	Tabbed Interface	5
4	User Interface	6
4.1	Layout	6
4.2	Device Panels	6
4.3	Responsive Design	6
4.4	Theme Support	6
5	Workflows	6
5.1	Monitoring a Single Battery Pack	6
5.2	Comparing Multiple Battery Packs	6
5.3	Debugging BLE Communication	6
6	Architecture	7
6.1	Architecture Overview	7
6.2	Class Diagram	8
6.3	Full System Diagram	9
6.4	Main BLE Connection Sequence	10
6.5	Data Processing Sequence	12
6.6	Application States	14
7	Configuration	15
7.1	BLE Protocol Constants	15
7.2	Protocol Variants	15
7.3	Heartbeat	15
8	Troubleshooting	15
8.1	Browser Compatibility Warning	15
8.2	BMS Not Found in Bluetooth Picker	15
8.3	Connection Drops	15
8.4	No Data After Connection	15
8.5	Incorrect Cell Count	16

1 Introduction

JK BMS Monitor is a browser-based application for monitoring JK Battery Management Systems via Web Bluetooth. It connects directly to JK BMS devices over Bluetooth Low Energy (BLE) and displays real-time battery telemetry including cell voltages, temperatures, current flow, state of charge, and protection status.

The application supports simultaneous monitoring of up to three BMS devices, with each device assigned a distinct color-coded slot (green, blue, orange). It implements the JK BMS BLE protocol as documented in the `esphome-jk-bms` project, supporting three protocol variants: JK02_24S, JK02_32S, and JK04.

Key capabilities:

- Real-time BLE communication with JK BMS devices
- Support for up to 32 cell monitoring per device
- Multi-device monitoring (up to 3 simultaneous connections)
- Cell voltage visualization with bar charts and statistics
- Temperature monitoring (MOS, battery probes, internal)
- MOSFET status monitoring (charge, discharge, balance)
- Raw hex data inspection for protocol debugging
- Activity logging per device
- PWA support for mobile use

2 Getting Started

2.1 System Requirements

- **Browser:** Google Chrome, Microsoft Edge, or Opera (Web Bluetooth API required)
- **Operating System:** Android, Windows, macOS, or Chrome OS (iOS/Safari do not support Web Bluetooth)
- **Hardware:** A JK BMS with Bluetooth capability

Web Bluetooth is **not** supported in Firefox or Safari. The app displays a warning banner if the browser lacks support.

2.2 Accessing the Application

JK BMS Monitor is available at:

<https://jk-bms-app.pages.dev/>

It can also be accessed through the BSI Portal at `blacksphereindustries.nl`.

2.3 Connecting a BMS

1. Ensure your JK BMS is powered on and Bluetooth is enabled
2. Open the app in a compatible browser
3. Click the “**Add BMS**” button
4. In the browser’s Bluetooth device picker, select your JK BMS
5. The app establishes a BLE connection and begins receiving telemetry data

3 Features

3.1 Multi-Device Monitoring

The app supports connecting up to three JK BMS devices simultaneously. Each connected device is assigned a color-coded slot:

- **Slot 0** (Green) — First connected device
- **Slot 1** (Blue) — Second connected device
- **Slot 2** (Orange) — Third connected device

When all three slots are occupied, the “Add BMS” card is hidden. Disconnecting a device frees its slot for a new connection.

3.2 Battery Overview

Each device panel displays at a glance:

- **Pack Voltage** — Total battery pack voltage in volts
- **Pack Current** — Current flow in amps (positive = charging, negative = discharging)
- **Pack Power** — Computed power in watts
- **State of Charge (SOC)** — Battery charge percentage with a color-gradient progress bar
- **State of Health (SOH)** — Battery health percentage
- **Remaining Capacity** — Current usable capacity in Ah
- **Full Capacity** — Total battery capacity in Ah
- **Charge Cycles** — Lifetime charge cycle count
- **Cycle Capacity** — Total energy cycled in Ah
- **Runtime** — Estimated remaining runtime

3.3 Cell Voltage Monitoring

The cell voltage section provides:

- **Cell Summary** — Shows minimum, average, and maximum cell voltages at a glance
- **Cell Bar Chart** — Each cell is displayed as a horizontal bar with:
 - Cell number label
 - Color-coded bar (red = low, yellow = marginal, green = OK, orange = high)
 - Exact voltage reading
- **Delta Voltage** — The difference between the highest and lowest cell, a key indicator of cell balance

3.4 Temperature Monitoring

Temperature readings are displayed in a grid layout:

- **MOS Temperature** — MOSFET transistor temperature
- **Battery Temp 1 & 2** — External battery probe temperatures
- **Internal Temp** — Internal BMS board temperature

Temperatures are color-coded:

- Green: Normal operating range
- Orange: Warm (approaching limits)
- Red: Hot (near or exceeding protection thresholds)

3.5 MOSFET and Protection Status

A three-column status grid shows:

- **Charge MOS** — On/Off status (green/red) of the charge MOSFET
- **Discharge MOS** — On/Off status of the discharge MOSFET
- **Balance** — Active/Inactive status of cell balancing

The balance current is also displayed when balancing is active.

3.6 Error Detection

The app decodes the BMS error bitmask and displays active faults:

- Charge/discharge overtemperature and undertemperature
- Cell/pack overvoltage and undervoltage
- Charge/discharge overcurrent and short circuit
- MOSFET overtemperature
- Cell count mismatch
- Current sensor anomaly
- Wire resistance fault
- Coprocessor communication error

3.7 Device Information

The Info tab displays:

- BMS model name
- Firmware version
- Hardware version
- Serial number
- Device name

3.8 Debug Tools

3.8.1 Hex Dump

The Debug tab provides a raw hex dump view of BLE frames with color-coded sections:

- Header bytes (amber)
- Data payload (white)
- CRC bytes (yellow)
- Offset addresses (gray)

3.8.2 Activity Log

A scrollable log records connection events, frame counts, protocol detection, and errors with color-coded severity:

- Info (amber)
- OK (green)
- Warning (yellow)
- Error (red)

3.9 Tabbed Interface

Each device panel uses tabs to organize information:

- **Overview** — Battery stats, cell voltages, temperatures
- **Info** — Device identification
- **Debug** — Hex dump and activity log

4 User Interface

4.1 Layout

The app uses a responsive grid layout:

- **Header** — Sticky top bar with app title, BLE icon, and global connection status
- **Device Grid** — Auto-filling grid of device panels (min 340px per card)
- **Add Card** — Dashed-border card for adding new BMS connections

4.2 Device Panels

Each connected device is rendered as a card with:

- **Panel Header** — Slot badge (colored), device name, connection status dot, and disconnect button
- **Panel Body** — Tabbed content area
- **Status Indicator** — Animated dot showing connection state:
 - Green with glow: Connected
 - Yellow with pulse: Connecting
 - Red: Disconnected

4.3 Responsive Design

On mobile screens (< 400px), the grid collapses to a single column and stat values are resized for readability. The app supports safe-area insets for notched mobile devices.

4.4 Theme Support

The app supports dark and light themes via CSS custom properties. When embedded in the BSI Portal, theme changes are relayed via `postMessage` using the BSI Theme Bridge pattern. In standalone mode, a theme toggle is available.

5 Workflows

5.1 Monitoring a Single Battery Pack

1. Open the app and click “Add BMS”
2. Select your JK BMS from the Bluetooth picker
3. The Overview tab appears with real-time data
4. Monitor cell voltages for balance issues (check delta voltage)
5. Watch temperatures during charge/discharge cycles
6. Check MOSFET status to verify charge/discharge is enabled

5.2 Comparing Multiple Battery Packs

1. Connect the first BMS (green slot)
2. Click “Add BMS” again for the second device (blue slot)
3. Optionally connect a third device (orange slot)
4. Compare cell voltage distributions across packs
5. Identify packs with poor cell balance (high delta voltage)

5.3 Debugging BLE Communication

1. Connect to a BMS device
2. Switch to the Debug tab
3. Observe the hex dump for incoming BLE frames
4. Check the activity log for protocol detection and frame counts

5. Use this information to diagnose communication issues

6 Architecture

The application is a single HTML file with all JavaScript and CSS inline. It uses the Web Bluetooth API for BLE communication and implements the JK BMS protocol parser.

6.1 Architecture Overview

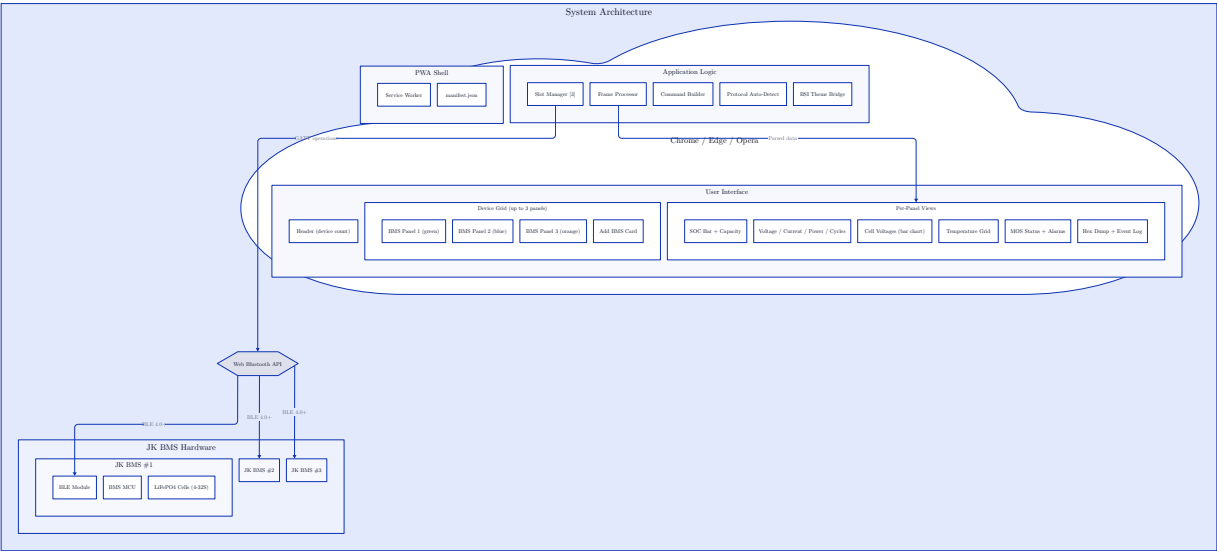


Figure 1: System architecture showing the BLE communication layer, protocol parser, and UI rendering.

6.2 Class Diagram

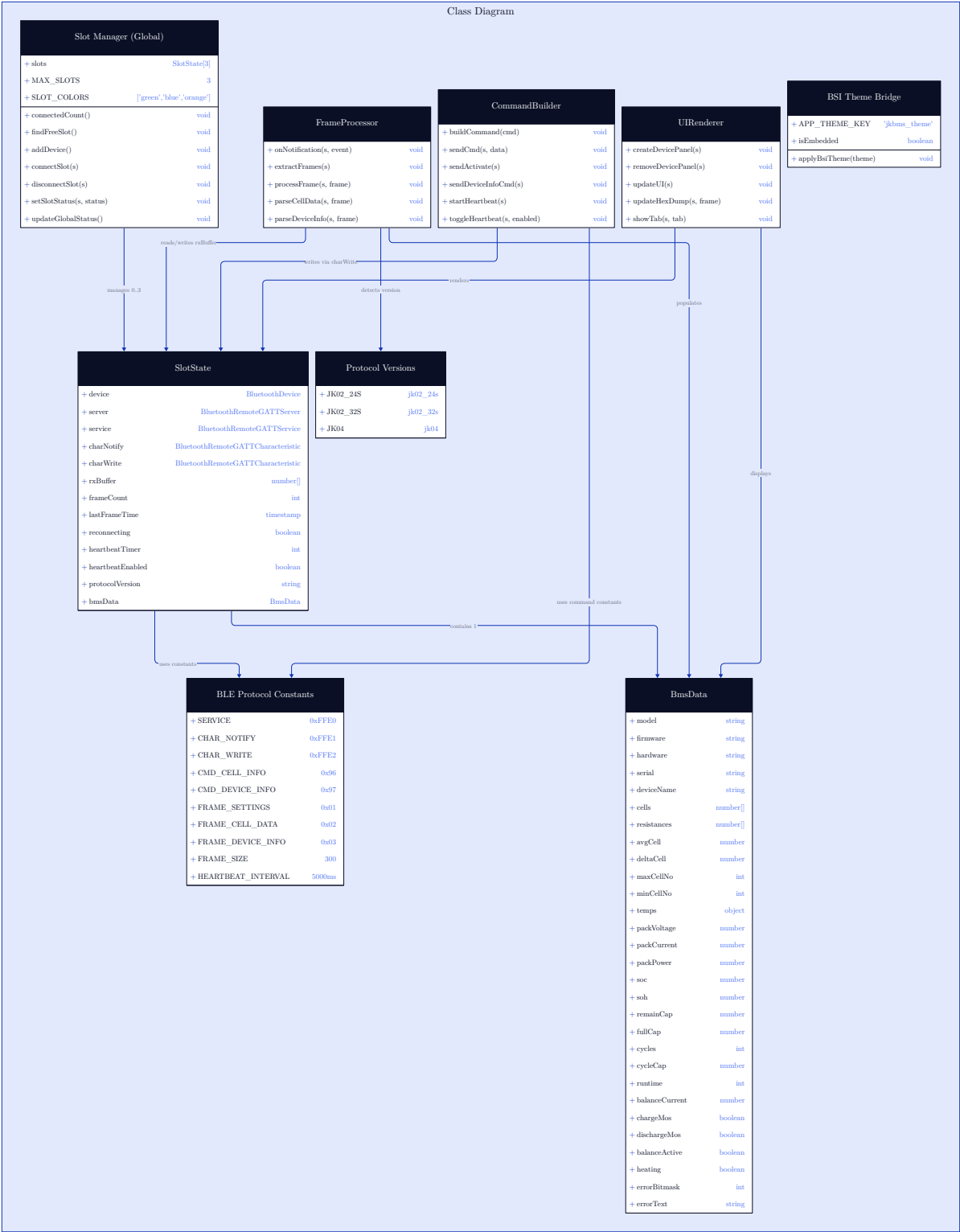


Figure 2: Class diagram showing the slot management system, BMS data model, and BLE protocol constants.

6.3 Full System Diagram

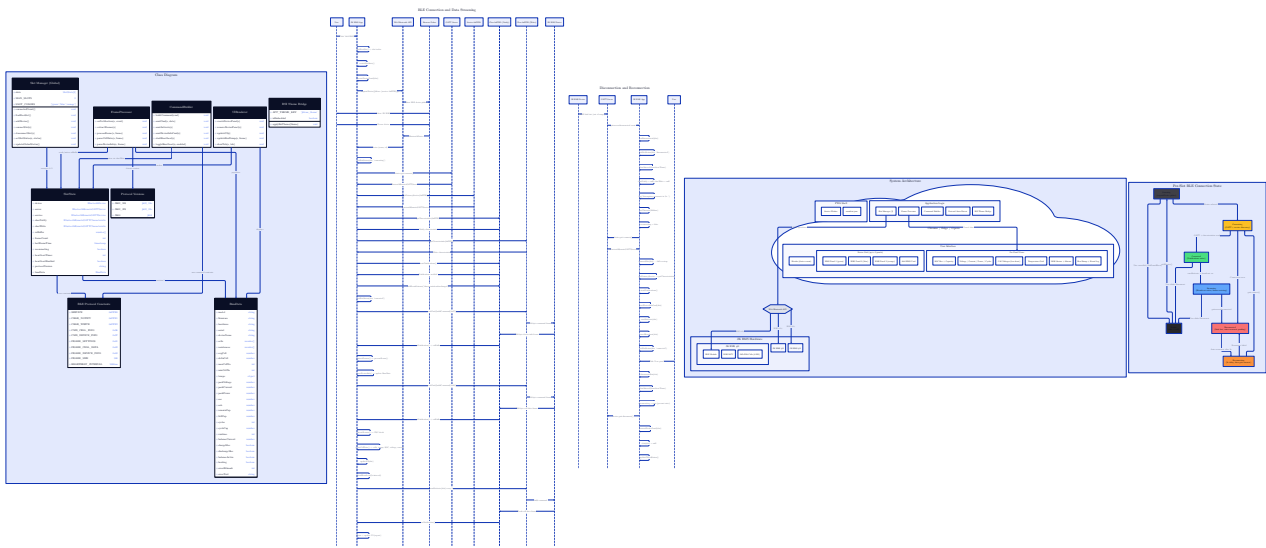


Figure 3: Comprehensive system diagram including BLE services, frame parsing, and multi-device state management.

6.4 Main BLE Connection Sequence

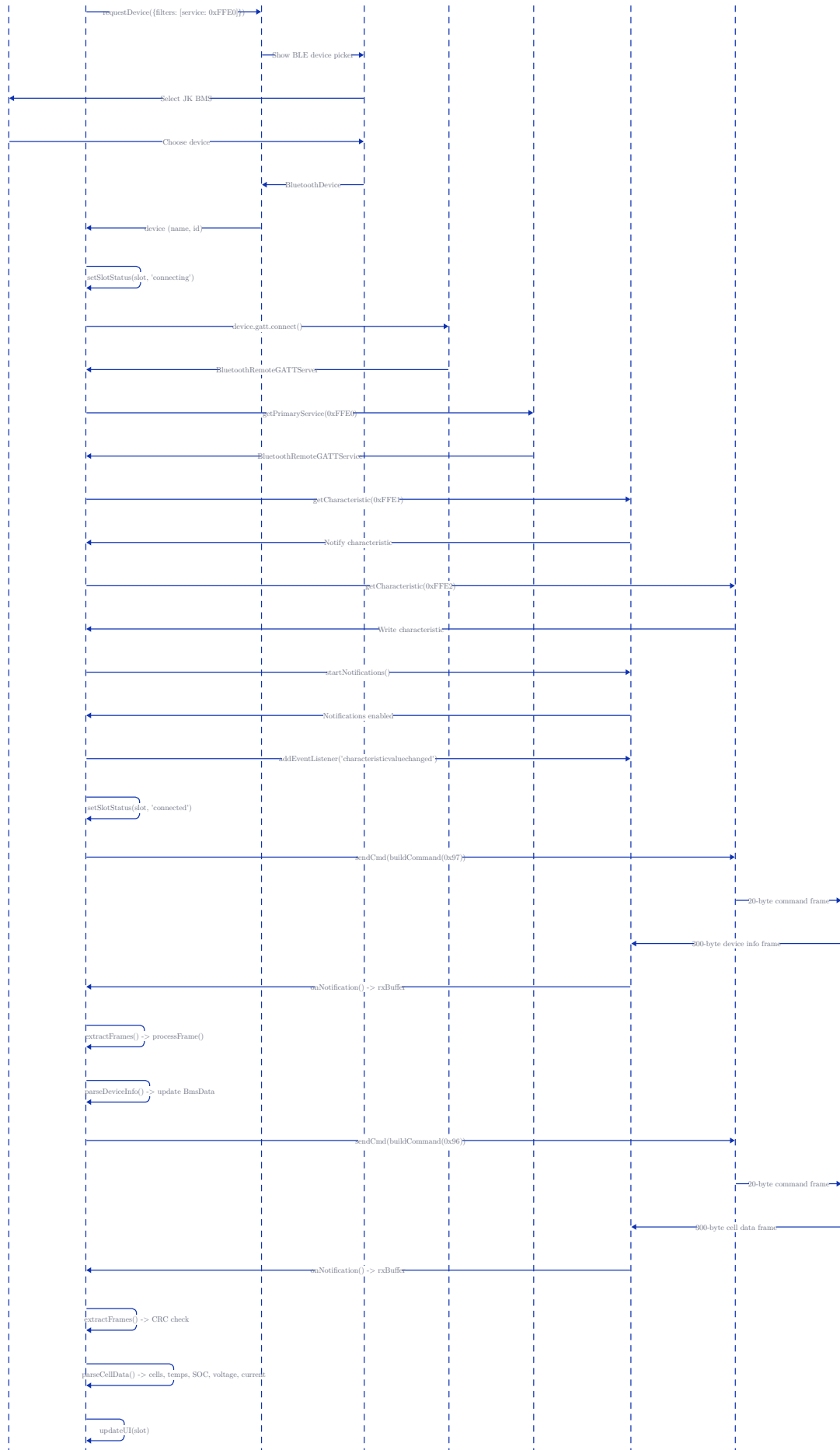


Figure 4: Sequence diagram showing the BLE connection flow from device discovery through GATT service setup to data streaming.

6.5 Data Processing Sequence

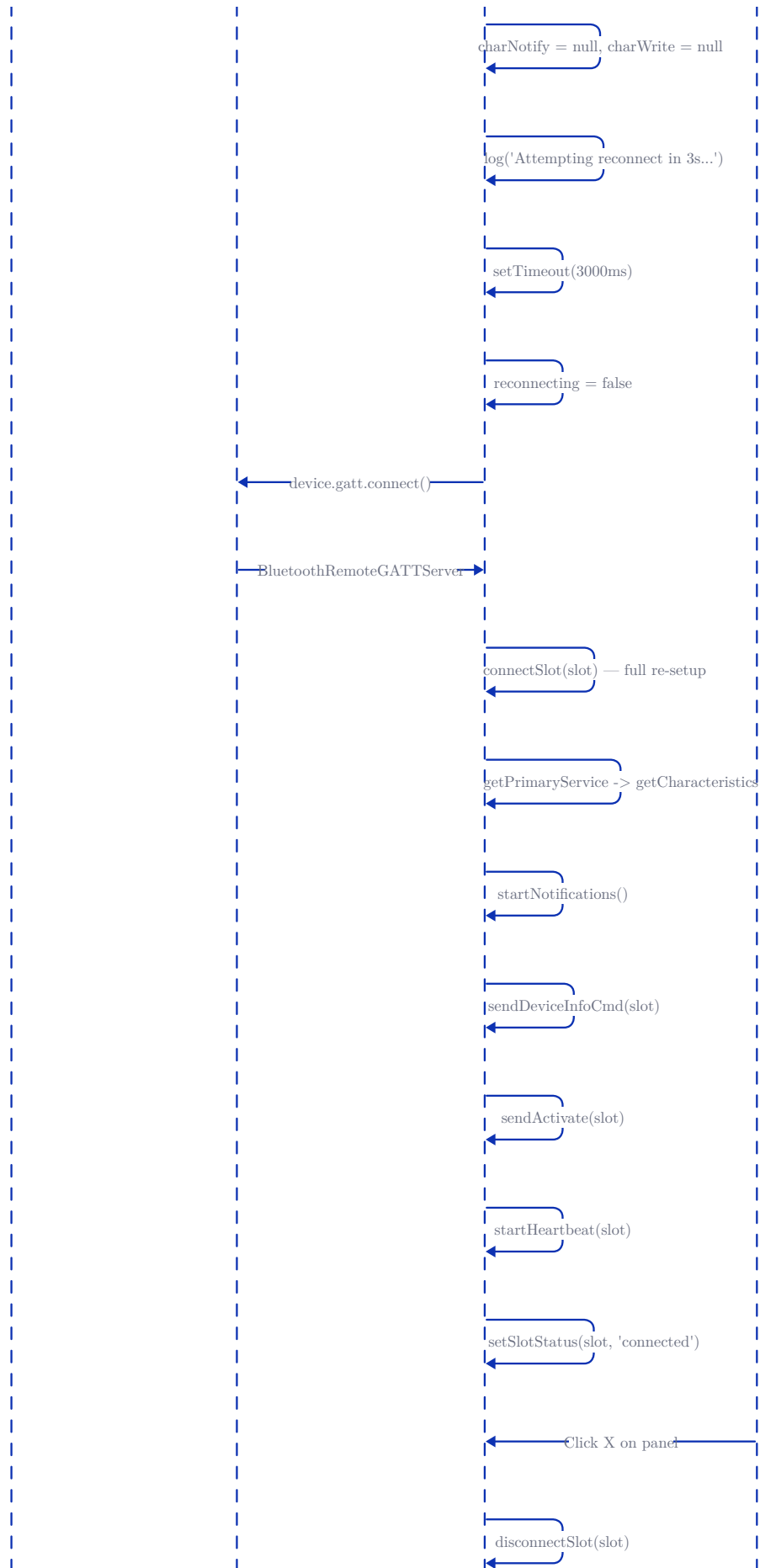


Figure 5: Sequence diagram showing how incoming BLE frames are buffered, parsed, and rendered to the UI.

6.6 Application States

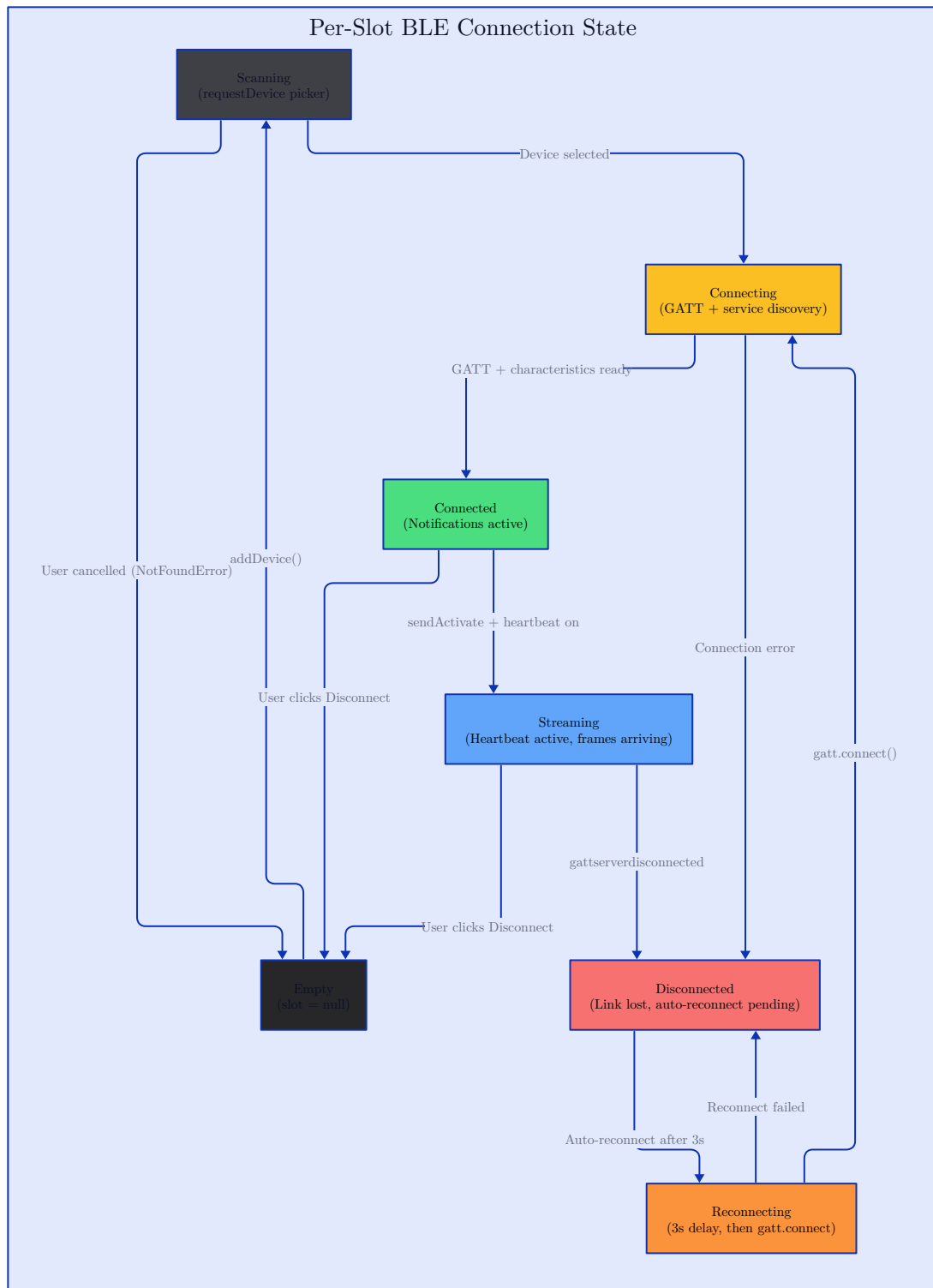


Figure 6: State diagram showing connection states (disconnected, connecting, connected, reconnecting) and slot lifecycle.

7 Configuration

7.1 BLE Protocol Constants

The application uses the following BLE service and characteristic UUIDs:

- Service UUID: 0xFFE0
- Notify Characteristic: 0xFFE1
- Write Characteristic: 0xFFE2

Command codes:

- Cell Info Request: 0x96
- Device Info Request: 0x97

Frame types:

- Settings: 0x01
- Cell Data: 0x02
- Device Info: 0x03

7.2 Protocol Variants

The app auto-detects the protocol variant based on frame structure:

- **JK02_24S** — For 24-cell JK02 series BMS
- **JK02_32S** — For 32-cell JK02 series BMS
- **JK04** — For JK04 series BMS

7.3 Heartbeat

A heartbeat signal is sent every 5 seconds (`HEARTBEAT_INTERVAL = 5000ms`) to maintain the BLE connection and request updated telemetry data.

8 Troubleshooting

8.1 Browser Compatibility Warning

If you see the yellow warning banner “Web Bluetooth is not supported,” you must switch to a compatible browser:

- **Supported:** Chrome (Android/Desktop), Edge, Opera
- **Not Supported:** Firefox, Safari, iOS browsers

8.2 BMS Not Found in Bluetooth Picker

- Ensure the BMS is powered on and not connected to another device
- Move closer to the BMS (BLE range is typically 5–10 meters)
- Try powering the BMS off and on again
- Check that Bluetooth is enabled on your computer/phone

8.3 Connection Drops

The app includes automatic reconnection logic. If a connection drops:

- The status dot turns red
- The app attempts to reconnect automatically
- Check the Debug log for error details
- Ensure the BMS is still powered and in range

8.4 No Data After Connection

- Switch to the Debug tab and check the hex dump for incoming frames

- Verify the frame counter is incrementing
- If no frames arrive, the BMS may require a firmware update
- Try disconnecting and reconnecting

8.5 Incorrect Cell Count

If the app shows fewer cells than expected, the protocol variant may be misdetected. Check the activity log for the detected protocol version. The app auto-detects between JK02_24S, JK02_32S, and JK04 based on frame structure.